

HR Database Sample Tables

Employee table

EmpId	Name	Dept	Salary	Location	JoinYear
E101	Alice	HR	55000	Delhi	2022
E102	Bob	HR	45000	Mumbai	2023
E103	Charlie	IT	65000	Delhi	2021
E104	Diana	IT	52000	Bangalore	2022
E105	Eve	Finance	70000	Mumbai	2020
E106	Frank	Finance	48000	Delhi	2023
E107	Grace	Sales	38000	Bangalore	2024
E108	Harry	Sales	42000	Mumbai	2022

Department table

Dept	DeptHead	Budget	Location	EstYear
HR	Priya	1200000	Delhi	2018
IT	Raj	2500000	Delhi	2019
Finance	Neha	1800000	Mumbai	2017
Sales	Amit	900000	Bangalore	2020
Marketing	Sara	1100000	Mumbai	2019

R&D	Vikram	2200000	Bangalore	2021
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SQL Setup Script

```
CREATE TABLE Employee (
    EmpId VARCHAR(10) PRIMARY KEY, Name VARCHAR(20), Dept VARCHAR(20),
    Salary INT, Location VARCHAR(20), JoinYear INT
);

INSERT INTO Employee VALUES
('E101', 'Alice', 'HR', 55000, 'Delhi', 2022),
('E102', 'Bob', 'HR', 45000, 'Mumbai', 2023),
('E103', 'Charlie', 'IT', 65000, 'Delhi', 2021),
('E104', 'Diana', 'IT', 52000, 'Bangalore', 2022),
('E105', 'Eve', 'Finance', 70000, 'Mumbai', 2020),
('E106', 'Frank', 'Finance', 48000, 'Delhi', 2023),
('E107', 'Grace', 'Sales', 38000, 'Bangalore', 2024),
('E108', 'Harry', 'Sales', 42000, 'Mumbai', 2022);

CREATE TABLE Department (
    Dept VARCHAR(20) PRIMARY KEY, DeptHead VARCHAR(20), Budget INT,
    Location VARCHAR(20), EstYear INT
);

INSERT INTO Department VALUES
('HR', 'Priya', 1200000, 'Delhi', 2018),
('IT', 'Raj', 2500000, 'Delhi', 2019),
('Finance', 'Neha', 1800000, 'Mumbai', 2017),
('Sales', 'Amit', 900000, 'Bangalore', 2020),
('Marketing', 'Sara', 1100000, 'Mumbai', 2019),
('R&D', 'Vikram', 2200000, 'Bangalore', 2021);
```

A1. Selection + Projection ($\sigma + \pi$)

List EmpId, Name, and Salary of all employees in IT or Finance who earn more than 50000.

```
SELECT EmpId, Name, Salary
FROM Employee
WHERE Dept IN ('IT', 'Finance') AND Salary > 50000;
```

Result:

EmpId	Name	Salary
E103	Charlie	65000
E105	Eve	70000

A2. Selection + Projection ($\sigma + \pi$)

List distinct Locations where employees joined in 2022 or later.

```
SELECT DISTINCT Location
FROM Employee
WHERE JoinYear >= 2022;
```

Result:

Location
Delhi
Mumbai
Bangalore

B1. Union ($\cup$)

List EmpId of employees who work either in Delhi or Mumbai.

```
SELECT EmpId FROM Employee WHERE Location = 'Delhi'
UNION
```

```
SELECT EmpId FROM Employee WHERE Location = 'Mumbai';
```

Result:

EmpId
E101
E102
E103
E105
E106
E108

B2. Intersection ($\cap$)

List EmpId of employees who are in departments with budget > 1500000 AND joined in 2022.

```
SELECT e.EmpId
FROM Employee e, Department d
WHERE e.Dept = d.Dept AND d.Budget > 1500000 AND e.JoinYear = 2022
INTERSECT
SELECT e.EmpId
FROM Employee e
WHERE e.JoinYear = 2022;
```

Result:

EmpId
E101
E104

B3. Set Difference ($-$)

List EmpId of employees in Sales but NOT in Bangalore.

```
SELECT EmpId FROM Employee WHERE Dept = 'Sales'
EXCEPT
SELECT EmpId FROM Employee WHERE Location = 'Bangalore';
```

Result:

EmpId
E108

C1. Cartesian Product + Selection ($\times + \sigma$)

Using explicit cross join, list Name and DeptHead for every employee paired with every department where employee's Location = department's Location.

```
SELECT e.Name, d.DeptHead
FROM Employee e, Department d
WHERE e.Location = d.Location;
```

Result:

Name	DeptHead
Alice	Priya
Alice	Raj
Charlie	Priya
Charlie	Raj
Frank	Priya
Frank	Raj
Bob	Neha
Eve	Neha
Harry	Neha

Diana	Amit
Grace	Amit
Vikram	Amit

C2. Cartesian Product + Non- Equi Selection ($\times + \sigma \neq$)

Using explicit cross join, list Name and DeptHead for employees whose Salary is GREATER THAN their department's EstYear (non- equi join).

```
SELECT e.Name, d.DeptHead
FROM Employee e, Department d
WHERE e.Dept = d.Dept AND e.Salary > d.EstYear;
```

Result:

Name	DeptHead
Alice	Priya
Bob	Priya
Charlie	Raj
Diana	Raj
Eve	Neha
Frank	Neha
Grace	Amit
Harry	Amit